



# higher education & training

Department:  
Higher Education and Training  
**REPUBLIC OF SOUTH AFRICA**

**NATIONAL CERTIFICATE (VOCATIONAL)**

**MATHEMATICS**

(Second paper)

**NQF LEVEL 2**

(10501042)

**12 November 2025 (Y-paper)**

**13:00–16:00**

Non-programmable scientific calculators may be used.

This question paper consists of 9 pages, 3 addenda and 1 formula sheet.

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**DEPARTMENT OF HIGHER EDUCATION AND TRAINING**  
**REPUBLIC OF SOUTH AFRICA**  
**NATIONAL CERTIFICATE (VOCATIONAL)**  
**MATHEMATICS**  
**NQF LEVEL 2**  
**(Second paper)**  
**TIME: 3 HOURS**  
**MARKS: 100**

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**INSTRUCTIONS AND INFORMATION**

1. Answer all the questions.
  2. Read all the questions carefully.
  3. Number the answers according to the numbering system used in this question paper.
  4. Clearly show all calculations, diagrams and graphs used in determining the answers.
  5. Round off answers to THREE decimals unless stated otherwise.
  6. Diagrams are not drawn to scale.
  7. Use only a black or blue pen.
  8. Write neatly and legibly.
-

2)

-3-

## QUESTION 1

- 1.1 A traffic police officer records the different brands of cars at a police checkpoint, as shown in the table below.

|          |          |        |          |         |          |          |          |
|----------|----------|--------|----------|---------|----------|----------|----------|
| Toyota   | Hyundai  | Toyota | Hyundai  | VW      | Toyota   | Mercedes | Toyota   |
| Toyota   | Mercedes | Ford   | Mercedes | Hyundai | Toyota   | Mercedes | Mercedes |
| Mercedes | VW       | Toyota | VW       | Toyota  | Ford     | Ford     | Toyota   |
| Ford     | Toyota   | VW     | Toyota   | VW      | Mercedes | Ford     | Ford     |
| Hyundai  | Toyota   | Toyota | Ford     | Toyota  | Mercedes | Toyota   | Ford     |

The distribution table of the different car brands is shown below:

| Car brand | Tally | Frequency |
|-----------|-------|-----------|
| Toyota    |       |           |
| VW        |       |           |
| Hyundai   |       |           |
| Mercedes  |       |           |
| Ford      |       |           |
| Total     |       | 40        |

- 1.1.1 Complete the distribution table on ADDENDUM A (attached). (5)
- 1.1.2 Which car brand is the most popular? (1)
- 1.1.3 Use the information from QUESTION 1.1.1 above to construct a bar chart on ADDENDUM A (attached). (4)

- 1.2 Use the data below to answer the following questions:

|    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|
| 72 | 63 | 23 | 71 | 72 | 84 | 76 | 42 | 53 | 13 | 37 |
|----|----|----|----|----|----|----|----|----|----|----|

- 1.2.1 Arrange the data in ascending order. (1)
- 1.2.2 Determine the middle quartile ( $Q_2$ ). (1)
- 1.2.3 Calculate the lower quartile ( $Q_1$ ). (2)
- 1.2.4 Calculate the lower quartile ( $Q_3$ ). (2)
- 1.2.5 Calculate the range. (1)
- 1.2.6 Calculate the interquartile range (IQR). (1)
- 1.2.7 Calculate the 85<sup>th</sup> percentile. (3)

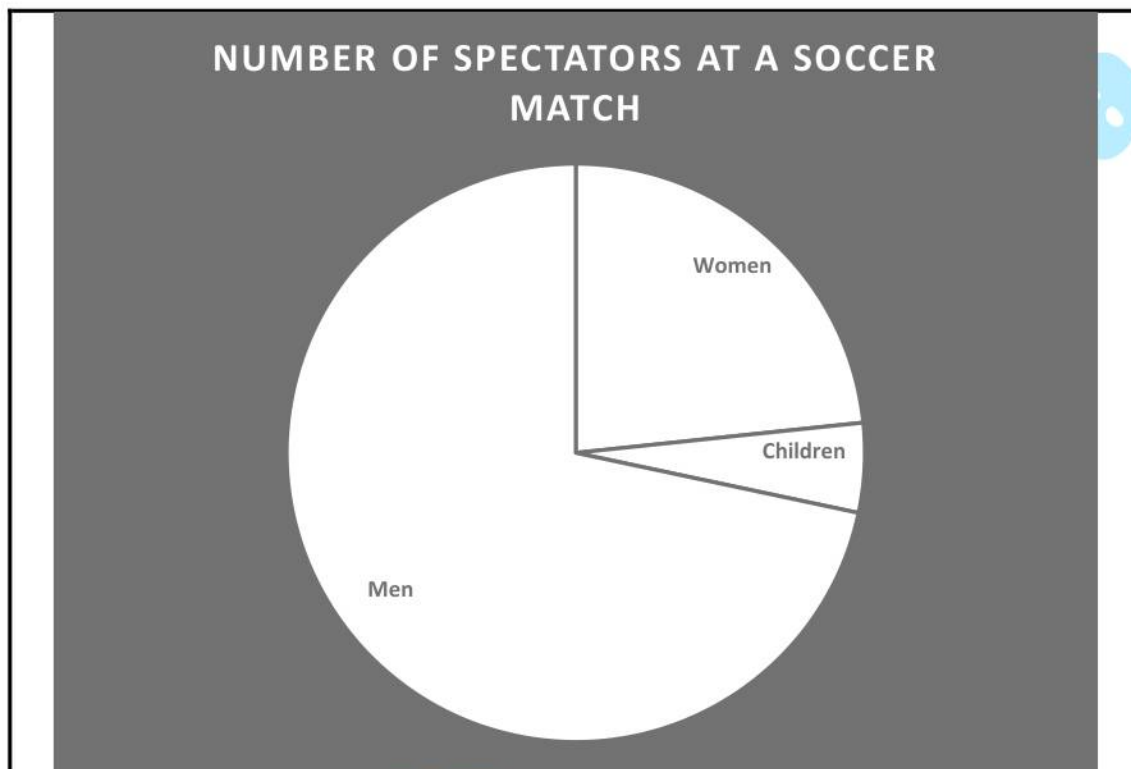
2)

-4-

1.3

The pie chart (not drawn to scale) below shows the proportion of men, women and children who attended a soccer match at the FNB stadium between Orlando Pirates and Kaizer Chiefs. Among the 90 000 spectators who attended the match, 4 500 were children and 21 000 were women.

Calculate the following:



1.3.1 The size (in degrees) of the sector for women.

1.3.2 The size (in degrees) of the sector for children.

1.3.3 The size (in degrees) of the sector for men.

(3 × 2) (6)

1.4

The table below represents the marks (out of 100) of 20 students in a Mathematics examination. It is also noted that the mode of the data is 32.

|    |    |    |    |    |
|----|----|----|----|----|
| 42 | 12 | 72 | 82 | 32 |
| 22 | 12 | 42 | 62 | 62 |
| 72 | M  | 22 | 52 | 42 |
| 32 | 22 | 62 | 32 | 52 |

1.4.1 Determine the value of M.

(1)

1.4.2 Draw a stem-and-leaf plot for the data.

(8)

1.4.3 Calculate the mean of the data.

(2)

2)

-5-

1.5

Define the following terminology as used in statistics:

1.5.1 Quartile

1.5.2 Percentile

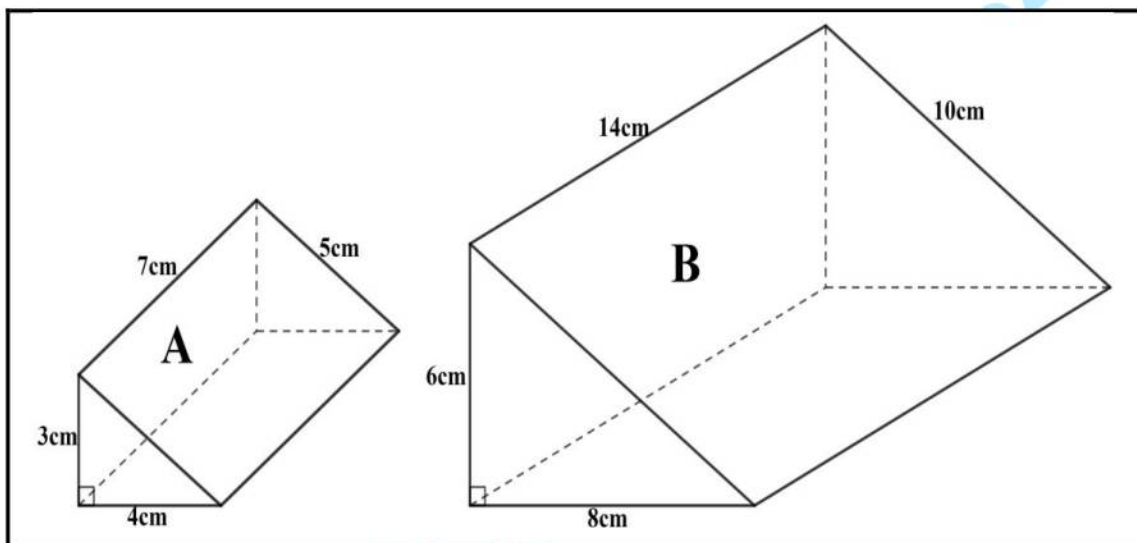
(2 × 1)

(2)

[40]

## QUESTION 2

2.1 Given below are two triangular prisms (not drawn to scale) labelled A and B.



2.1.1 Calculate the surface area ( $S_A$ ) of prism A. (2)

2.1.2 Calculate the surface area ( $S_B$ ) of prism B. (2)

2.1.3 Determine and simplify the ratio  $S_A : S_B$ . (2)

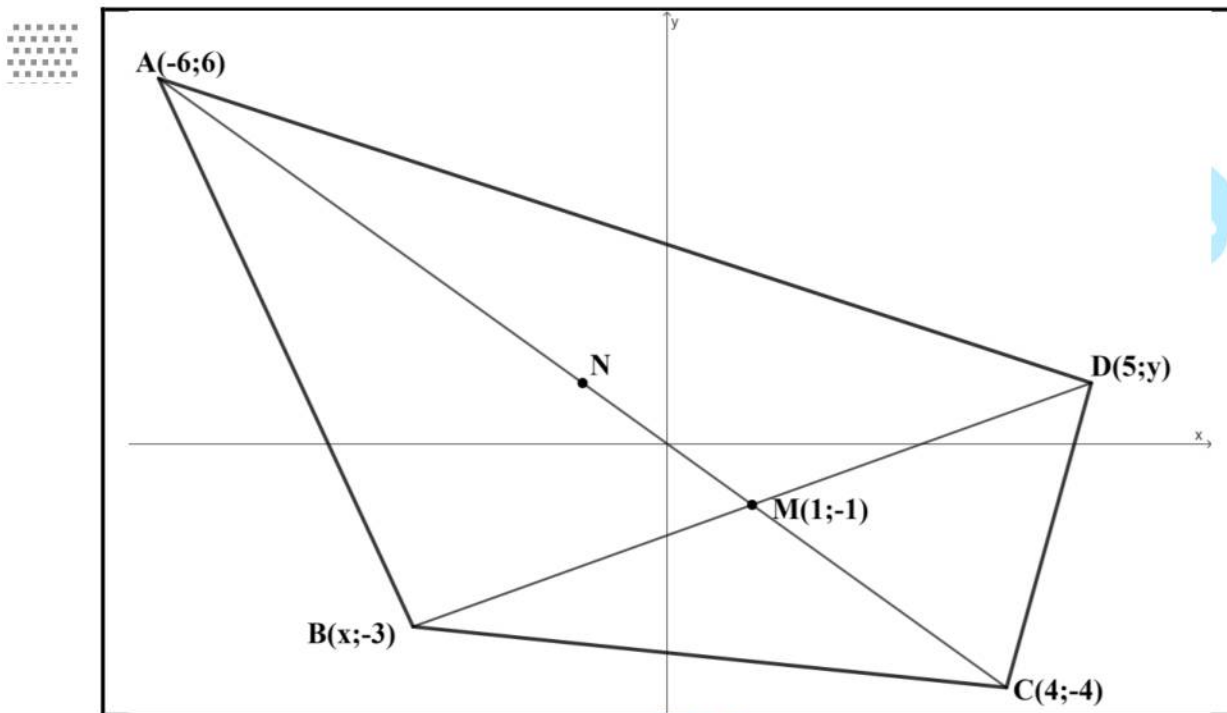
2.1.4 What conclusion can you draw from your answer to QUESTION 2.1.3 above? (1)

2.)

-6-

2.2

The diagram below is a quadrilateral with vertices  $A(-6;6)$ ;  $B(x;-3)$ ;  $C(4;-4)$  and  $D(5;y)$ . Given that  $M(1;-1)$  is the midpoint of  $BD$  while  $N$  is the midpoint of  $AC$ , calculate the following:



- 2.2.1 The coordinates of  $N$ . (2)
- 2.2.2 The value of  $y$ . (2)
- 2.2.3 The value of  $x$ . (2)
- 2.2.4 The length of  $AB$ . Leave your answer in surd form. (3)
- 2.2.5 The gradient of  $CD$ . (2)

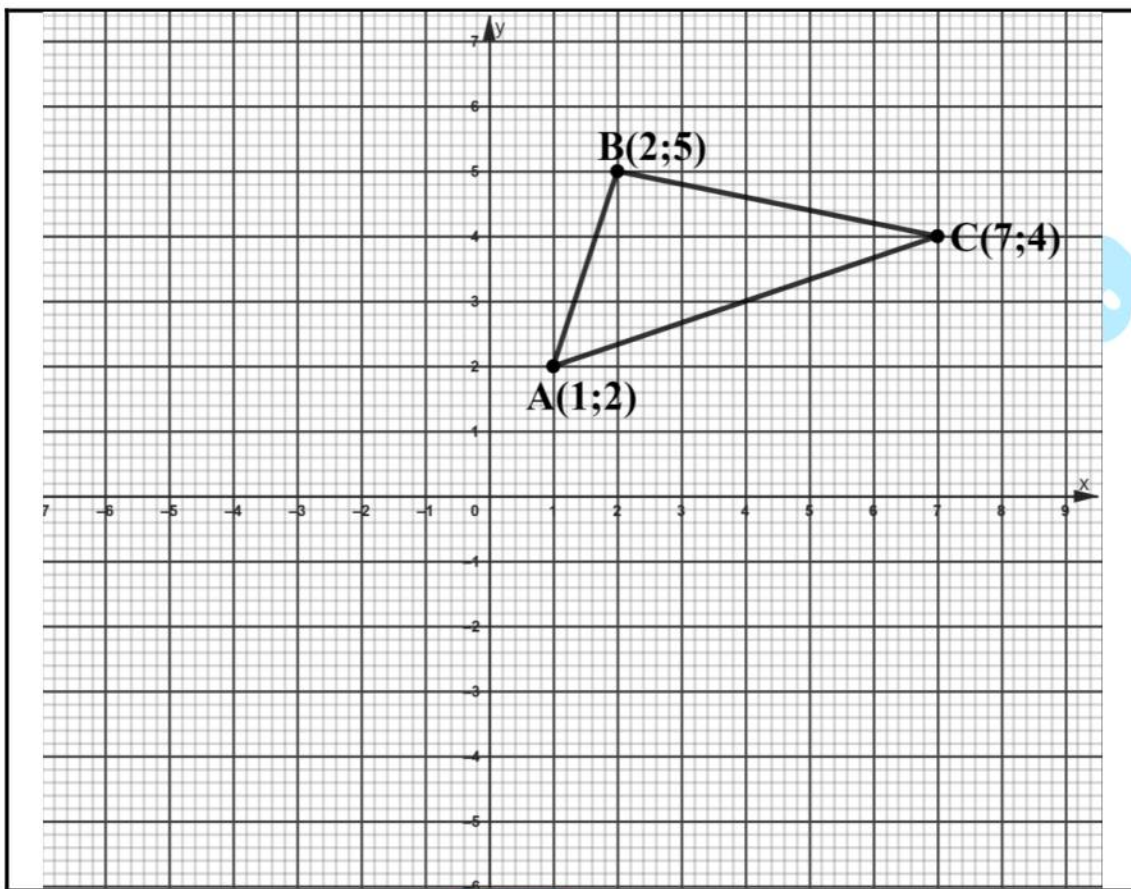


2.)

-7-

2.3

Consider  $\triangle ABC$  in the diagram below.



2.3.1 Draw the image  $\triangle A'B'C'$  of  $\triangle ABC$  after a reflection about the y-axis on ADDENDUM B (attached).

2.3.2 Draw the image  $\triangle A''B''C''$  of  $\triangle ABC$  after a reflection about the x-axis on ADDENDUM B (attached).

2.3.3 Draw the image  $\triangle A'''B'''C'''$  of  $\triangle ABC$  after a translation of 6 units downwards and 7 units to the left on ADDENDUM B (attached).

2.3.4 Write down the coordinates of  $\triangle PQR$  formed by translating  $\triangle ABC$  about the line  $x = y$ .

(4 × 3)

(12)  
[30]

2)

-8-

### QUESTION 3

3.1 Define the following terminology used in trigonometry:

3.1.1 Period

3.1.2 Amplitude

(2 × 1) (2)

3.2 Given  $f(x) = 2\cos x + 2$  and  $g(x) = 2\sin x$  for  $0^\circ \leq x \leq 360^\circ$ .

3.2.1 Calculate the values of A, B, C, D, E, P, Q, R, S and T in the table below.

|        |   |     |     |    |     |     |     |     |      |     |      |     |     |
|--------|---|-----|-----|----|-----|-----|-----|-----|------|-----|------|-----|-----|
| $x$    | 0 | 30  | 60  | 90 | 120 | 150 | 180 | 210 | 240  | 270 | 300  | 330 | 360 |
| $f(x)$ | 4 | 3,7 | A   | 2  | B   | 0,3 | C   | 0,3 | 1    | D   | 3    | 3,7 | E   |
| $g(x)$ | 0 | P   | 1,7 | Q  | 1,7 | 1   | 0   | R   | -1,7 | S   | -1,7 | T   | 0   |

(2)

3.2.2 Make use of ADDENDUM C (attached) and the table in QUESTION 3.2.1 above to draw the graphs of  $f(x)$  and  $g(x)$ .

(4)

3.2.3 Determine  $A_1 + A_2$  where  $A_1 = \text{Amplitude of } f(x)$  and  $A_2 = \text{Amplitude of } g(x)$ .

(2)

3.2.4 Read from the graph in QUESTION 3.2.2 above, the values of  $x$  for which  $f(x) = 3$ .

(2)

3.3 Given that  $\tan \theta = \frac{6}{8}$  for  $180^\circ \leq \theta \leq 270^\circ$ :

3.3.1 Represent the information on a Cartesian plane using a triangle.

(1)

3.3.2 Calculate the hypotenuse of the triangle in QUESTION 3.3.1 above.

(2)

3.3.3 Calculate the value of  $3\sin \theta - \cos \theta$ .

(3)

3.3.4 Calculate the value of  $2\sin \theta \cos \theta - \frac{2}{\tan \theta}$ .

(3)

3.4 Solve for  $\theta$  in the following where  $0^\circ \leq \theta \leq 90^\circ$ .

$$\sin(\theta - 12^\circ) = \frac{1}{2}$$

(2)

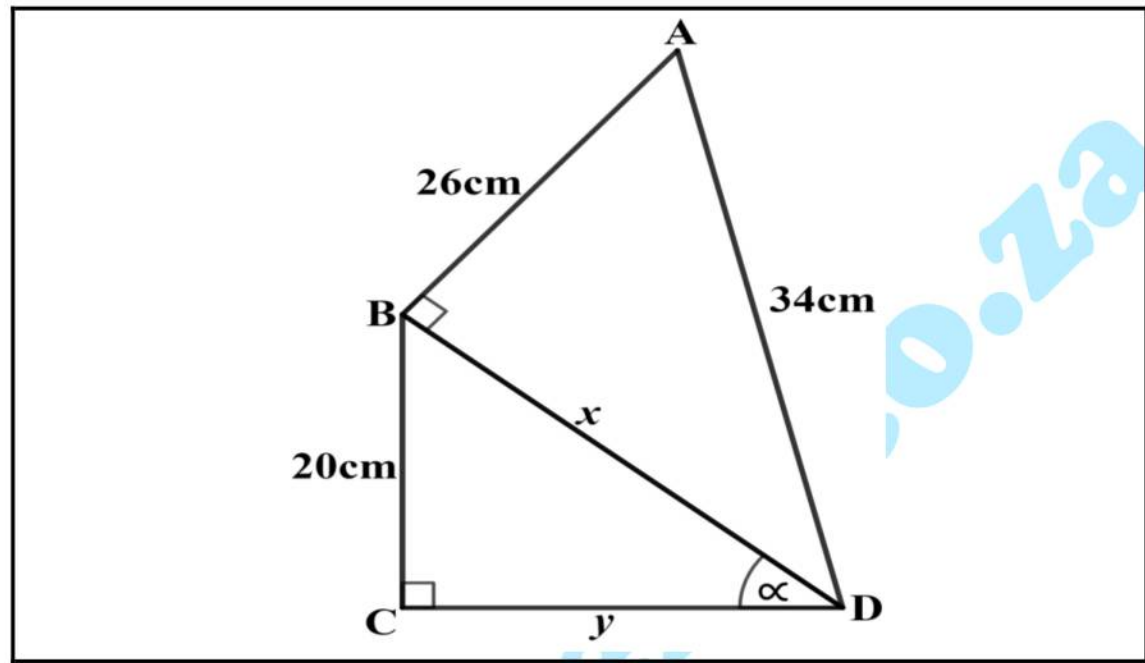
2)

-9-

3.5

Consider the diagram below where  $AD = 34$  cm;  $CD = y$ ;  $BC = 20$  cm;  
 $AB = 26$  cm;  $BD = x$  and  $\widehat{BDC} = \alpha$ .

Calculate the following:



- 3.5.1 The value of  $x$ . (2)
- 3.5.2 The value of  $y$ . (2)
- 3.5.3 The value of  $\alpha$ . (3)

[30]



**TOTAL: 100**

2)

-1-

**ADDENDUM A**

**EXAMINATION NUMBER:**

|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
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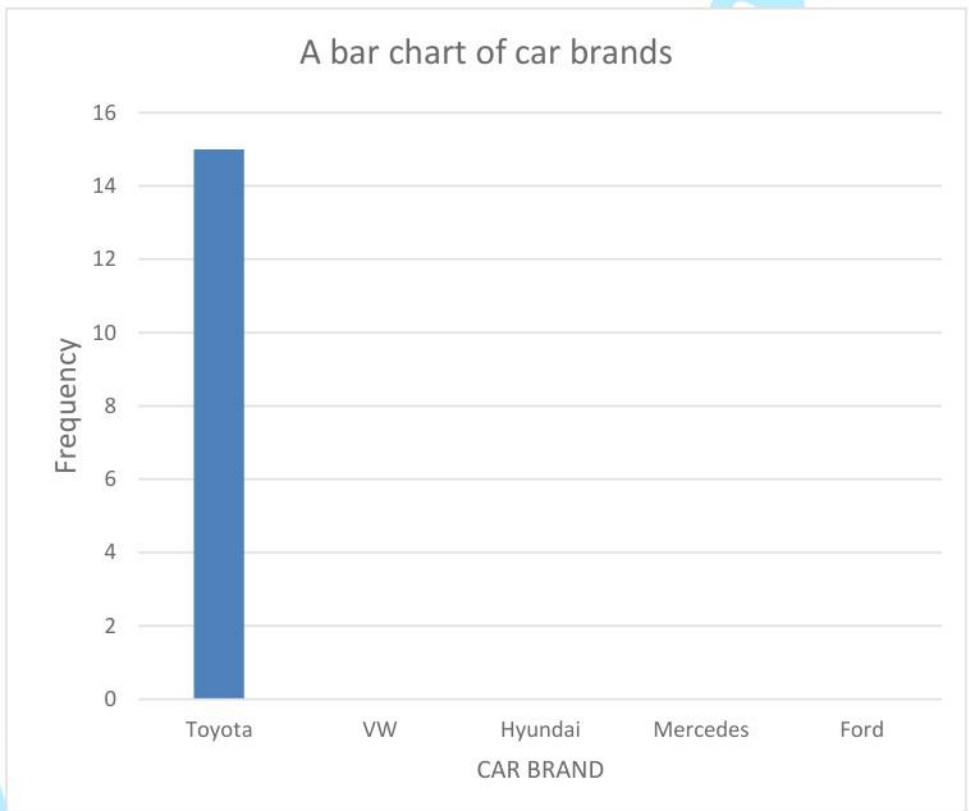
**QUESTION 1**

1.1

1.1.1

| Car brand | Tally | Frequency |
|-----------|-------|-----------|
| Toyota    |       |           |
| VW        |       |           |
| Hyundai   |       |           |
| Mercedes  |       |           |
| Ford      |       |           |
|           | Total | 40        |

1.1.3



2)

-2-

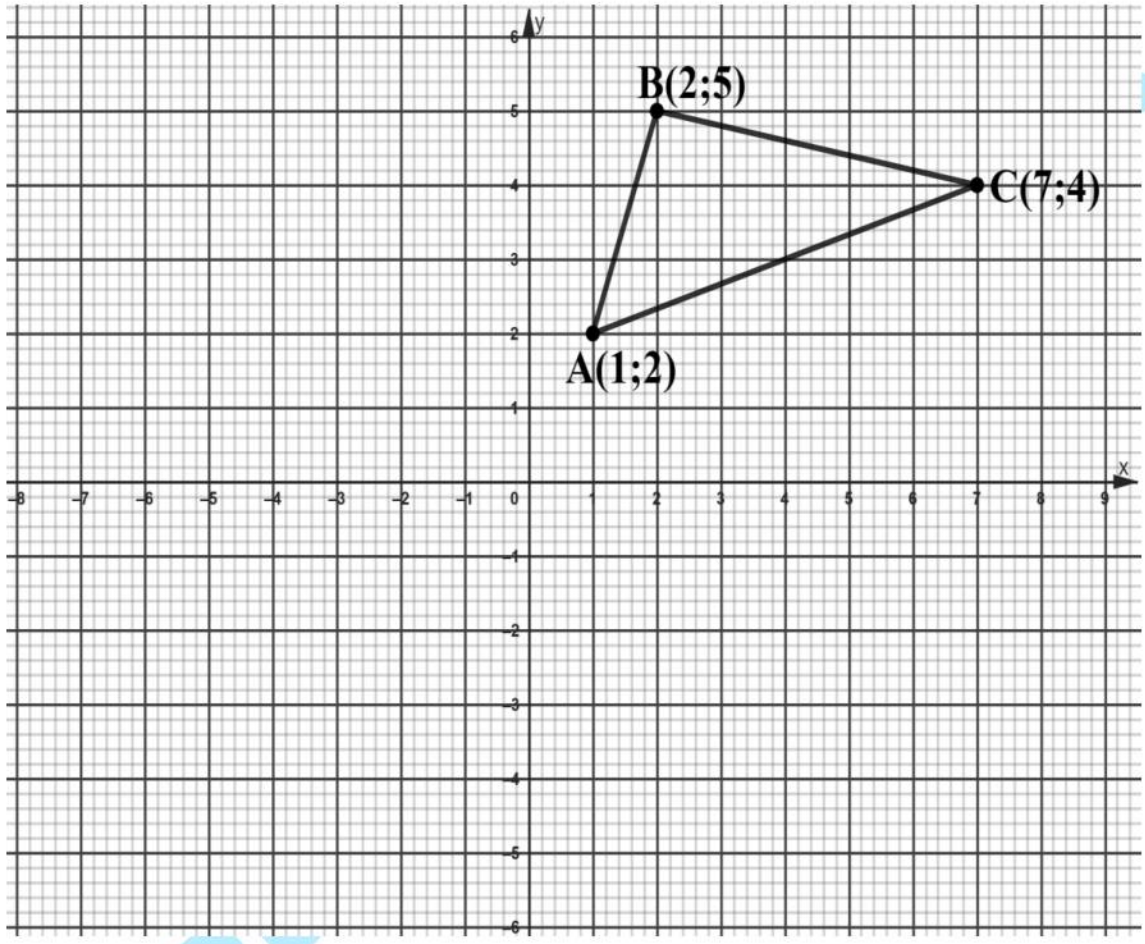
ADDENDUM B

EXAMINATION NUMBER:

|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
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|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|

**QUESTION 2**

2.3 2.3.1; 2.3.2 & 2.3.3



2.)

-3-

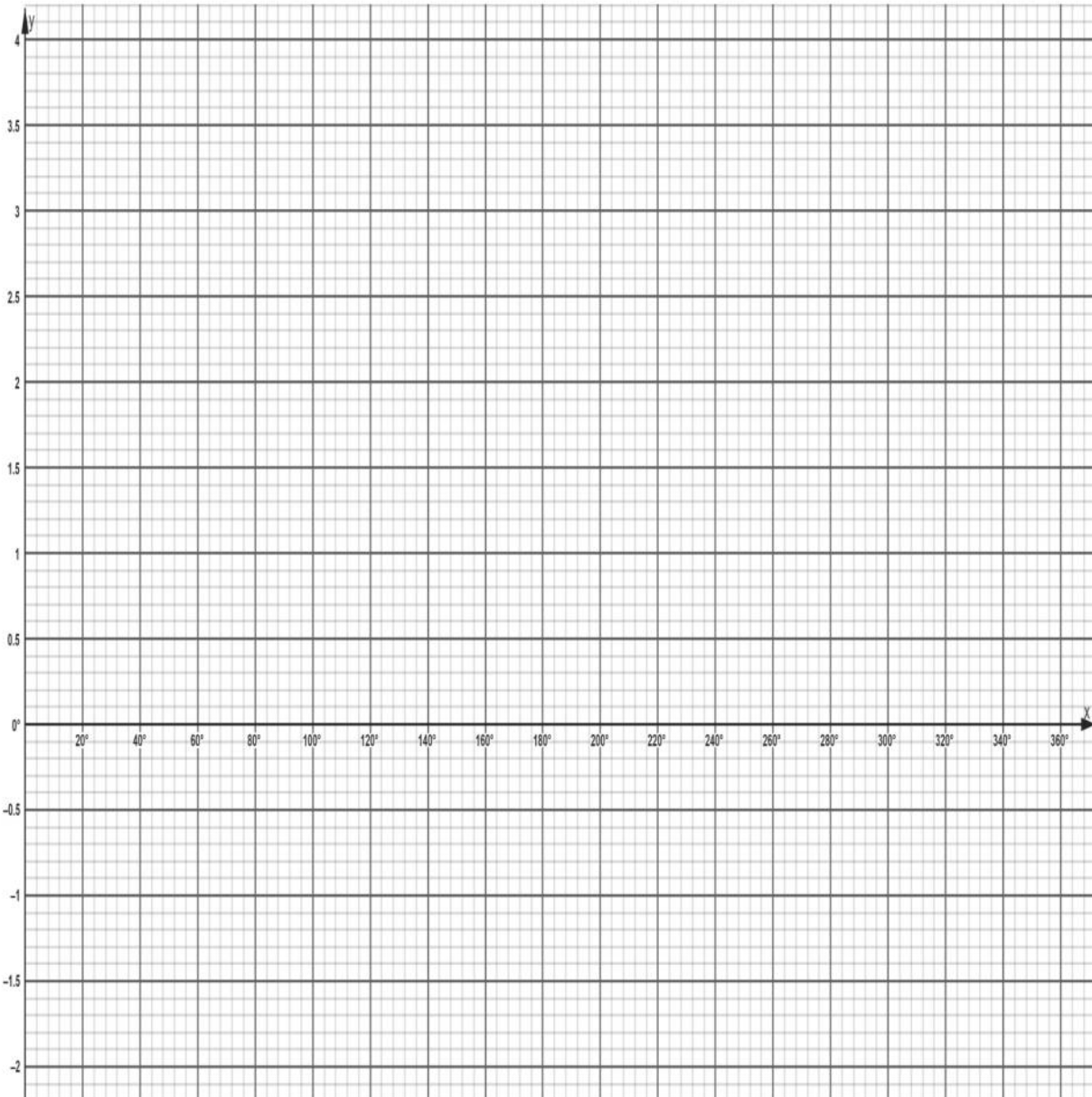
ADDENDUM C

EXAMINATION NUMBER:

|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
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|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|

QUESTION 3

3.2 3.2.2



2)

## TULA SHEET

1.  $A_{\text{square}} = l \times l = l^2$
2.  $A_{\text{rectangle}} = l \times w$
3.  $A_{\text{triangle}} = \frac{1}{2} b \times h$
4.  $A_{\text{circle}} = \pi r^2$
5.  $C = 2\pi r$
6.  $\text{Area of parallelogram} = \text{base} \times \text{perpendicular height}$
7.  $A_{\text{hexagon}} = \frac{3\sqrt{3}}{2} L^2$
8.  $A_{\text{hexagon}} = \frac{\sqrt{3}}{2} W^2$
9.  $A_{\text{cylinder}} = 2\pi r(h + r)$
10.  $\text{Volume} = \text{Area of base} \times \text{perpendicular height}$
11.  $V_{\text{cylinder}} = \pi r^2 h$
12.  $\text{Total surface area of a triangular prism}$   
 $= (\text{height of prism} \times \text{perimeter of base}) + 2(\text{area of base})$
13.  $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$
14.  $D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
15.  $M = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
16.  $\theta = \tan^{-1} m$
17.  $\bar{x} = \sum \frac{f_i x_i}{n}$  OR  $\text{Mean} = (\text{total or sum of all items}) \div (\text{number of items})$
18.  $R = X_n - X_1$  OR  $\text{Range} = \text{highest value} - \text{lowest value}$
19.  $Q_{j\text{position}} = \frac{j}{4}(n + 1)$
20.  $\text{Interquartile range} = \text{upper quartile} - \text{lower quartile} = (Q_3 - Q_1)$
21.  $\text{Semi - interquartile range} = \frac{1}{2}(\text{upper quartile} - \text{lower quartile}) = \frac{Q_3 - Q_1}{2}$
22.  $P_{j\text{position}} = \frac{j}{100}(n + 1)$